

End-of-Module-Exams 2024

Signature: _____

I hereby confirm that all the pages of my exam paper
have been stapled to this cover sheet.

Number of supplementary answer sheets attached:

Total number of added pages:

Module: w.BA.XX.3DA-WIN.XX / Data Analytics
Exam date / time: 20.01.2025 / 14:00 - 15:00
Exam duration: 60 Minutes
Total number of pages: 23 (incl. cover sheet)
Resources allowed: Open book
Calculator: Approved calculator according to 'Guidelines on Supplementary Materials'
Dictionary: Yes

General information:

- Use a black or blue pen (indelible ink) to write your answers.
- Write your name on every page.
- Only write on the front side of each page.
- Supplementary answer sheets are provided by the invigilators. These must be clearly numbered and the number of supplementary answer sheets used must be noted on the cover sheet. Do not use your own sheets of paper.
- Sign the cover sheet before you hand in the exam.
- If you encounter an exam question that does not seem clear to you, you may feel the need to make assumptions. In such a case, summarize your assumptions in your answer.
- For remarks pertaining to specific questions, please refer to information included in the task.
- Toilet breaks are only allowed if the examination exceeds 90 minutes.
- Invigilators are not allowed to respond to any questions related to the content of the exam.
- If you wish to hand in your exam early, you may do so until 20 minutes before the official end of the examination.

Marked by:

Total points:

Grade:

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Last Name:

First Name:

Answer Sheet – Please note that only the answers on this sheet will be assessed.

Grading Table

Performance Assessment	Weighting	Grade	Weighted Grade
Performance Assessment 1 Written Exam	70%		
Performance Assessment 2 Project Work (Group Work)	30%		
Performance Assessment 3 Submission of the solutions of at least 10 weekly exercises	Pass/Fail	Pass/Fail	-
		(Grade on cover sheet)	
Module grade (Evento)			↔ Rounded ↔

Marking Table

No.	Task	Points	Corrector's, initials
1	Single-Choice Questions		
Total Points			
Grade			

Overview

Task	Description	Points
1	Single-Choice Questions	60

TOTAL: 60 points

Last Name:

First Name:

Answer-Sheet (please mark the correct answer per question):

Question	a)	b)	c)	d)
Q1	☐	☐	☐	☐
Q2	☐	☐	☐	☐
Q3	☐	☐	☐	☐
Q4	☐	☐	☐	☐
Q5	☐	☐	☐	☐
Q6	☐	☐	☐	☐
Q7	☐	☐	☐	☐
Q8	☐	☐	☐	☐
Q9	☐	☐	☐	☐
Q10	☐	☐	☐	☐
Q11	☐	☐	☐	☐
Q12	☐	☐	☐	☐
Q13	☐	☐	☐	☐
Q14	☐	☐	☐	☐
Q15	☐	☐	☐	☐

Question	a)	b)	c)	d)
Q16	☐	☐	☐	☐
Q17	☐	☐	☐	☐
Q18	☐	☐	☐	☐
Q19	☐	☐	☐	☐
Q20	☐	☐	☐	☐
Q21	☐	☐	☐	☐
Q22	☐	☐	☐	☐
Q23	☐	☐	☐	☐
Q24	☐	☐	☐	☐
Q25	☐	☐	☐	☐
Q26	☐	☐	☐	☐
Q27	☐	☐	☐	☐
Q28	☐	☐	☐	☐
Q29	☐	☐	☐	☐
Q30	☐	☐	☐	☐

Question Section – Please note that this question paper will not be assessed.

General Information

- This exam consists of **single-answer questions only**.
- This means there is only one correct answer for each question.
- The possible answers for each question are listed as (a), (b), (c), and (d).
- If you provide more than one answer for a question, you will receive no marks for that question.
- If you do not answer a question, you will receive no marks for that question.
- Use the answer sheet for your solutions (**answer section** of the exam). This page belongs to the **question section** of the exam.

Overview

Task	Description	Points
1	Single-Answer Questions	60

TOTAL: 60 points

Single-Answer Questions

(60 points)

Categorization of Data & Data Collection Methods																																																								
Q1: Consider the following table containing car data. Which of the statements with regard to the “scale of measure” of the variables is <u>not</u> correct? <table><tr><th></th><th>Brand</th><th>Model</th><th>Year</th><th>Engine_Size</th><th>Fuel_Type</th><th>Price_Category</th><th>Price</th></tr><tr><td>0</td><td>Toyota</td><td>Corolla</td><td>2020</td><td>1.8</td><td>Petrol</td><td>Mid</td><td>20000</td></tr><tr><td>1</td><td>Ford</td><td>Mustang</td><td>2021</td><td>5.0</td><td>Petrol</td><td>High</td><td>55000</td></tr><tr><td>2</td><td>BMW</td><td>X5</td><td>2019</td><td>3.0</td><td>Diesel</td><td>High</td><td>60000</td></tr><tr><td>3</td><td>Audi</td><td>A4</td><td>2022</td><td>2.0</td><td>Petrol</td><td>High</td><td>45000</td></tr><tr><td>4</td><td>Tesla</td><td>Model 3</td><td>2021</td><td>Electric</td><td>Electric</td><td>High</td><td>50000</td></tr></table> a) The variable <i>Model</i> has unordered category labels and is, therefore, nominal scaled. b) The variable <i>Price</i> has an absolute (or true) zero and is, therefore, ratio scaled. c) The variable <i>Fuel_Type</i> has ordered category labels and is, therefore, interval scaled. d) The variable <i>Price_Category</i> has ordered category labels and is, therefore, ordinal scaled.									Brand	Model	Year	Engine_Size	Fuel_Type	Price_Category	Price	0	Toyota	Corolla	2020	1.8	Petrol	Mid	20000	1	Ford	Mustang	2021	5.0	Petrol	High	55000	2	BMW	X5	2019	3.0	Diesel	High	60000	3	Audi	A4	2022	2.0	Petrol	High	45000	4	Tesla	Model 3	2021	Electric	Electric	High	50000	2 points
	Brand	Model	Year	Engine_Size	Fuel_Type	Price_Category	Price																																																	
0	Toyota	Corolla	2020	1.8	Petrol	Mid	20000																																																	
1	Ford	Mustang	2021	5.0	Petrol	High	55000																																																	
2	BMW	X5	2019	3.0	Diesel	High	60000																																																	
3	Audi	A4	2022	2.0	Petrol	High	45000																																																	
4	Tesla	Model 3	2021	Electric	Electric	High	50000																																																	
Q2: Which of the statements with regard to the following five-point Likert scale from a questionnaire is <u>not</u> correct? Note the different levels which range from <i>Strongly Agree</i> to <i>Strongly Disagree</i>. <table><tr><th colspan="6">5 Point Likert Scale</th></tr><tr><th>Numbering</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th></tr><tr><td></td><td>Strongly Agree</td><td>Agree</td><td>Neutral</td><td>Disagree</td><td>Strongly Disagree</td></tr><tr><td>I like Stock Market</td><td>☐</td><td>☐</td><td>☐</td><td>☐</td><td>☐</td></tr><tr><td>I like Stocks</td><td>☐</td><td>☐</td><td>☐</td><td>☐</td><td>☐</td></tr><tr><td>I Like Money</td><td>☐</td><td>☐</td><td>☐</td><td>☐</td><td>☐</td></tr><tr><td>I Like Return</td><td>☐</td><td>☐</td><td>☐</td><td>☐</td><td>☐</td></tr><tr><td>Total</td><td>4</td><td>8</td><td>12</td><td>16</td><td>20</td></tr></table> a) Likert scale data are ordinal data in which levels are often represented by numbers. b) Likert scale data requires special statistical methods (e.g., rank-based correlations). c) For Likert scale data, the distance between levels is generally not meaningful. d) Likert scale data are always treated as interval data.								5 Point Likert Scale						Numbering	1	2	3	4	5		Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree	I like Stock Market	☐	☐	☐	☐	☐	I like Stocks	☐	☐	☐	☐	☐	I Like Money	☐	☐	☐	☐	☐	I Like Return	☐	☐	☐	☐	☐	Total	4	8	12	16	20	2 points
5 Point Likert Scale																																																								
Numbering	1	2	3	4	5																																																			
	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree																																																			
I like Stock Market	☐	☐	☐	☐	☐																																																			
I like Stocks	☐	☐	☐	☐	☐																																																			
I Like Money	☐	☐	☐	☐	☐																																																			
I Like Return	☐	☐	☐	☐	☐																																																			
Total	4	8	12	16	20																																																			

Q3: The text item "MacBook Air" must be extracted from the following HTML file using an XPath. Which is the correct XPath for extracting this text item? <pre><!DOCTYPE html> <html> <head> <title>Product Page</title> </head> <body> <div id="main"> <h1>Our Latest Devices</h1> <p>Discover the cutting-edge technology we offer.</p> <li class="product">MacBook Air <li class="product">Dell XPS 13 <li class="product">HP Spectre x360 <footer> <p>Get in touch: support@techstore.com</p> </footer> </div> </body> </html></pre> a) /html/head/body/div/ul/li[2]/text() b) /html/body/div/ul/li[1]/text() c) /html/head/body/div/ul/li[1]/text() d) /html/body/div/ul/li[2]/text()	2 points
Data Types, Data Structures, & File Formats in Python	
Q4: Which data types are returned by the print() function in the code below? <pre>x = str(bool(int('0'))) y = 7.5 * 2 z = 'Python' + 'is easy' + 'to learn' w = True is True v = 100 >= 101 u = isinstance('hello', str) h = -15 print(type(x), type(y), type(z), type(w), type(v), type(u), type(h))</pre> a) str, int, str, True, False, True, int b) str, float, str, bool, bool, bool, int c) bool, float, str, bool, bool, str, int d) str, float, str, True, False, True, int	2 points

Q5: Which types are returned by the print() function in the code below?

2 points

```
model_01 = ('iPhone 12', 'Galaxy S21', 'iPhone 12', 'Galaxy S21')
model_02 = {'OnePlus', 'Pixel', 'OnePlus', 'Pixel'}
model_03 = ['Galaxy S21', 'iPhone 12', 'Galaxy S21', 'iPhone 12']
model_04 = {'iPhone 12': 2, 'Galaxy S21': 2}

print(type(model_01), type(model_02), type(model_03), type(model_04))
```

- a) set, list, dict, tuple
- b) tuple, set, list, dict
- c) list, dict, tuple, set
- d) dict, tuple, set, list

Q6: Consider the pandas data frame below named "smartphones" and the Python code. Which columns of the data frame will be returned by the print() function and which values? Remember that .loc is used to access rows and columns by column names, and .iloc is used to access rows and columns by their numerical indices. Also remember that != is the "not equal" (also called "not equal to") operator.

2 points

Data frame "smartphones":

	brand	model	release_year	price	battery_life
0	Samsung	Galaxy S20	2020	999	18
1	Apple	iPhone 12	2020	799	17
2	Xiaomi	Mi 11	2021	749	20
3	OnePlus	OnePlus 8	2020	699	19
4	Google	Pixel 5	2020	699	18
5	Huawei	P40	2020	899	16
6	Samsung	Galaxy S21	2021	1099	22
7	Apple	iPhone 13	2021	799	19

Python code:

```
# Subset
import pandas as pd
print(smartphones.loc[(smartphones['brand'] != 'Google') &
                      (smartphones['price'] <= 800) &
                      (smartphones['battery_life'] >= 19)].iloc[0, [0,1,2]])
```

- a) The columns 'brand', 'price,' and 'release_year' with the values "Google", 699, and "2020"
- b) The columns 'brand', 'model,' and 'release_year' with the values "Apple", "iPhone 12," and "2020"
- c) The columns 'brand', 'price,' and 'battery_life' with the values "Samsung", 999, and 18
- d) The columns 'brand', 'model,' and 'release_year' with the values "Xiaomi", "Mi 11," and "2021"

Q7: Imagine you must read data to a pandas data frame. The pandas library has already been imported as 'pd'. The name of the file is 'my_file'. You know that the file is 'utf-8' encoded but there is no file extension. You open the file with a text editor and look at its structure (see below). Which line of code reads the data correctly?

2 points

```
# Structure of the file named 'my_file'.
[{
  "type": "node",
  "id": 39947904,
  "lat": 47.3767316,
  "lon": 8.5421608,
  "tags": {
    "addr:city": "Zürich",
    "addr:country": "CH",
    "addr:housenumber": "1",
    "addr:postcode": "8001",
    "addr:street": "Bahnhofbrücke",
    "brand": "Coop",
    "brand:wikidata": "Q432564",
    "brand:wikipedia": "de:Coop (Schweiz)",
    "check_date": "2022-06-25",
    "check_date:opening_hours": "2022-09-22",
    "name": "Coop",
    "opening_hours": "Mo-Sa 06:00-22:00",
    "operator": "Coop Genossenschaft",
    "shop": "supermarket",
    "wheelchair": "yes"
  }
}]
```

- a) data = pd.read_csv("my_file", encoding="utf-8")
- b) data = pd.read_json("my_file", encoding="utf-8")
- c) data = pd.read_xml("my_file", encoding="utf-8")
- d) data = pd.read_html("my_file", encoding="utf-8")

Q8: Which statement with regard to the data and format below is not correct?

2 points

1.)

```
<?xml version="1.0" encoding="UTF-8"?>
<note>
  <to>Tove</to>
  <from>Jani</from>
  <heading>Reminder</heading>
  <body>Don't forget me this weekend!</body>
</note>
```

2.)

```
{
  "type": "node",
  "id": 39768209,
  "lat": 47.2250689,
  "lon": 8.9699805,
  "tags": {
    "addr:city": "Uznach",
    "addr:postcode": "8730",
    "alt_name": "Coop Linthpark",
    "branch": "Uznach Linthpark",
    "brand": "Coop",
    "name": "Coop",
  }
}
```

- a) The data shown under 1.) is organized in tags with start and end tags.
- b) The data shown under 2.) is referred to as semi-structured.
- c) The data shown under 1.) is referred to as unstructured.
- d) The data shown under 2.) is organized in name/value (also called key/value) pairs.

Data Preparation & Combining & Organizing Data

Q9: What is the output of the print() function in the following Python code?

2 points

```
import pandas as pd

# Creating the DataFrame for car data
data = {
    'make': ['VW', 'BMW', 'BMW', 'VW', 'BMW', 'VW'],
    'price': [21000, 27000, 33000, 31000, 45000, 32000]
}

df = pd.DataFrame(data)

output = df.groupby(by='make').price.mean()

# Convert the output to include the make
output_dict = output.to_dict()

print(output_dict)
```

- a) {'BMW': 34000.0, 'VW': 27000.0}
- b) {'BMW': 36000.0, 'VW': 26000.0}
- c) {'BMW': 35000.0, 'VW': 28000.0}
- d) {'BMW': 32000.0, 'VW': 24000.0}

Q10: What is the output of the print() function in the following Python code?

2 points

```
import pandas as pd

# Creating a DataFrame for smartphone data
data = {
    'Brand': ['Apple', 'Samsung', 'Apple', 'OnePlus', 'Samsung'],
    'Model': ['iPhone 12', 'Galaxy S21', 'iPhone 12', 'Nord', 'Galaxy S21'],
    'Price': [999, 850, 999, 500, 960],
    'Sales': [1500, 2000, 1800, 1200, 1600]
}

# Creating the DataFrame
smartphones = pd.DataFrame(data)

# Creating a pivot table to summarize the values
pivot_smartphones = pd.pivot_table(smartphones,
                                    index=['Brand', 'Model'],
                                    values=['Price', 'Sales'],
                                    aggfunc='mean')

print(pivot_smartphones)
```

a)

		Price	Sales
Brand	Model		
Apple	iPhone 12	999.0	1650.0
OnePlus	Nord	500.0	1200.0
Samsung	Galaxy S21	905.0	1800.0

b)

		Price	Sales
Brand	Model		
Apple	iPhone 12	887.0	1650.0
OnePlus	Nord	500.0	1200.0
Samsung	Galaxy S21	905.0	2200.0

c)

		Price	Sales
Brand	Model		
Apple	iPhone 12	984.0	1650.0
OnePlus	Nord	500.0	1200.0
Samsung	Galaxy S21	905.0	1850.0

d)

		Price	Sales
Brand	Model		
Apple	iPhone 12	925.0	1650.0
OnePlus	Nord	500.0	1200.0
Samsung	Galaxy S21	905.0	1780.0

Q11: What is the output of the print() function in the following Python code?

2 points

```
import pandas as pd

# Creating the first DataFrame with basic smartphone information
data_phones = {
    'Brand': ['Apple', 'Samsung', 'Apple', 'OnePlus'],
    'Model': ['iPhone 12', 'Galaxy S21', 'iPhone 13', 'Nord']
}

df_phones = pd.DataFrame(data_phones)

# Creating the second DataFrame with pricing and sales information
data_sales = {
    'Model': ['iPhone 12', 'Galaxy S21', 'iPhone 13', 'Nord'],
    'Price': [999, 850, 1099, 500],
    'Sales': [1500, 2000, 1700, 1200]
}

df_sales = pd.DataFrame(data_sales)

# Merging the two DataFrames on the 'Model' column
merged_df = pd.merge(df_phones, df_sales, on='Model')

# Display the merged DataFrame
print(merged_df)
```

a)

	Brand	Model	Price	Sales
0	Apple	iPhone 12	999	1500
1	Samsung	Galaxy S21	850	2000
2	Apple	iPhone 13	1099	1700
3	OnePlus	Nord	500	1200
4	Apple	iPhone 14	1999	2500

b)

	Brand	Model	Sales
0	Apple	iPhone 12	1500
1	Samsung	Galaxy S21	2000
2	Apple	iPhone 13	1700
3	OnePlus	Nord	1200

c)

	Model	Price	Sales
0	iPhone 12	999	1500
1	iPhone 13	1099	1700
2	Nord	500	1200

d)

	Brand	Model	Price	Sales
0	Apple	iPhone 12	999	1500
1	Samsung	Galaxy S21	850	2000
2	Apple	iPhone 13	1099	1700
3	OnePlus	Nord	500	1200

Q12: What is the output of the print() function in the following Python code?

2 points

```
import pandas as pd
import re

# Creating a small DataFrame with two columns
data = {
    'product': ['Phone A', 'Phone B', 'Phone C', 'Phone D', 'Phone E'],
    'price_raw': ['$999', '€850', '1099 USD', '500 pounds', 'Free']
}

df = pd.DataFrame(data)

# Extract values
price = []
for i in df['price_raw']:
    d1 = re.findall('[0-9]+', i)
    try:
        d2 = d1[0].strip()
    except IndexError:
        d2 = None
    price.append(d2)

# Save as a new variable
df['price'] = pd.Series(price, dtype="float64")

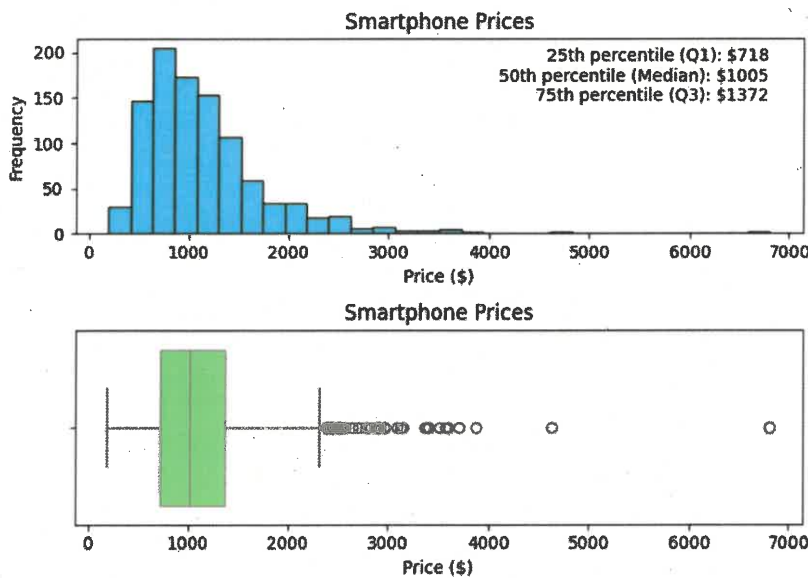
# Print value
print(list(df['price']))
```

- a) [999, 850, 1099, 500, None]
- b) [999.0, 850.0, 1099.0, 500.0, nan]
- c) ['\$', '€', 'USD', 'pounds', '']
- d) [999.0, 850.0, 1099.0, 500.0, False]

Exploratory Data Analysis (EDA)

Q13: Consider the following graphics showing the distribution of smartphone prices (in U.S. \$) and the measures of central tendency of the price distribution. Question: Which statement with regard to these graphics and measures is not correct?

2 points



- a) The interquartile range (IQR) of smartphone prices is \$545.
- b) The calculation \$1372 minus \$718 provides the spread of the middle 50% of smartphone prices which is also shown in the box of the boxplot.
- c) The smartphone prices have a right-skewed (positively skewed) distribution.
- d) It can be concluded from the graphs and data that the mean smartphone price is > \$1005.

Q14: Consider the following contingency table. It contains data from an online survey. Which statement with regard to this contingency table is not correct?

2 points

	Men	Women	Sum
I like dogs	50	70	120
I do not like dogs	20	60	80
Sum	70	130	200

- a) 60% of all respondents like dogs.
- b) 30% of all respondents are women who don't like dogs.
- c) 25% of all respondents are men who like dogs.
- d) 45% of all respondents don't like dogs.

Q15: Which interpretation of the correlation matrix below based on smartphone data is not correct? A description of variables is provided below the correlation matrix. Note that the statistical significance of correlations is not considered here.

2 points

	Price (USD)	Camera (megapixel)	Battery Life (hours)	Weight (grams)	Portability (rating)
Price (USD)	1.0000	0.9651	0.9073	0.5311	-0.9948
Camera (megapixel)	0.9651	1.0000	0.9025	0.6324	-0.9701
Battery Life (hours)	0.9073	0.9025	1.0000	0.6301	-0.9303
Weight (grams)	0.5311	0.6324	0.6301	1.0000	-0.6096
Portability (rating)	-0.9948	-0.9701	-0.9303	-0.6096	1.0000

Explanation of the variables used in the data set:

- Price (USD): Cost of the smartphone in U.S. dollars.
- Camera (megapixel): Camera resolution in megapixels.
- Battery Life (hours): Duration of battery life in hours.
- Weight (grams): Weight of the smartphone in grams.
- Portability (rating): Portability rating on a 1-10 scale (higher values indicate more portable).

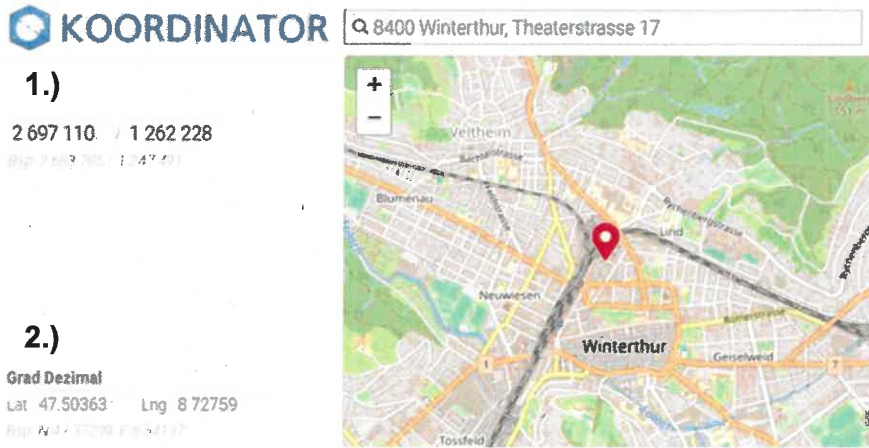
- a) The higher the price, the higher the camera resolution.
- b) More expensive smartphones tend to have a shorter battery life.
- c) The longer the battery life, the less portable the smartphone.
- d) Heavier smartphones tend to be less portable.

Q16: Consider the following data showing a sample of smartphones with their prices. Which is the correct value of the "sample standard deviation" of smartphone prices in this sample? Note that the values in the answer options are rounded to two decimal places.

2 points

Smartphone	Price
iPhone 13	800
Samsung Galaxy S21	950
Google Pixel 6	1200
OnePlus 9	850
Sony Xperia 5	750

- a) 184.36
- b) 205.16
- c) 178.19
- d) 169.84

Representation & Analysis of Geographical Data	
Q17: Look at the following map and the coordinates. Note that “Lat” stands for latitude and “Lng” stands for longitude. Also note the parts in the figure marked as 1.) and 2.). Which statement regarding the following map and coordinates is <u>not</u> correct?  1.) 2 697 110 / 1 262 228 Bsp 2 697 110 / 1 262 228 2.) Grad Dezimal Lat 47.50363 Lng 8.72759 Bsp 47.50363 / 8.72759 a) The coordinates given under 1.) refer to a geographic coordinate system. b) The coordinates given under 1.) refer to one of the Swiss coordinate systems. c) The coordinates given under 2.) refer to an internationally used coordinate system. d) The term “Grad Dezimal” (engl.: “degrees decimal”) given under 2.) refers to the World Geodetic System 1984 (WGS84).	2 points
Q18: Which statement regarding the following GIS data format is <u>not</u> correct?  <pre data-bbox="287 1473 868 1590">{ "type": "Polygon", "coordinates": [[[30, 10], [40, 40], [20, 40], [10, 20], [30, 10]]] }</pre> a) This data format is used to store vector data. b) The coordinates [30,10], [40,40], ... refer to points that form a polygon. c) This data format can also be used to store point and line geometries. d) This data format is referred to as Well Known Text (WKT).	2 points

Statistical Hypothesis Testing

Q19: Consider the following contingency table. It contains data from an online survey. For a Chi-squared test, the expected frequencies must be calculated. Which are the correct expected frequencies?

2 points

	Men	Women	Sum
I like dogs	50	70	120
I do not like dogs	20	60	80
Sum	70	130	200

a)

Expected frequencies:

[[48, 72],

[82, 64]]

b)

Expected frequencies:

[[42, 78],

[28, 52]]

c)

Expected frequencies:

[[52, 76],

[32, 51]]

d)

Expected frequencies:

[[62, 80],

[38, 42]]

Q20: Consider the data below (left table) representing the weights of watermelons on two different fields and the results of an unpaired two-sample t-test (right table). The null hypothesis is that the mean weight of watermelons in Field A does not differ from the mean weight of watermelons in Field B (two-sided hypothesis test). The significance level α was set to 5%. Which statement with regard to the t-test is not correct?

2 points

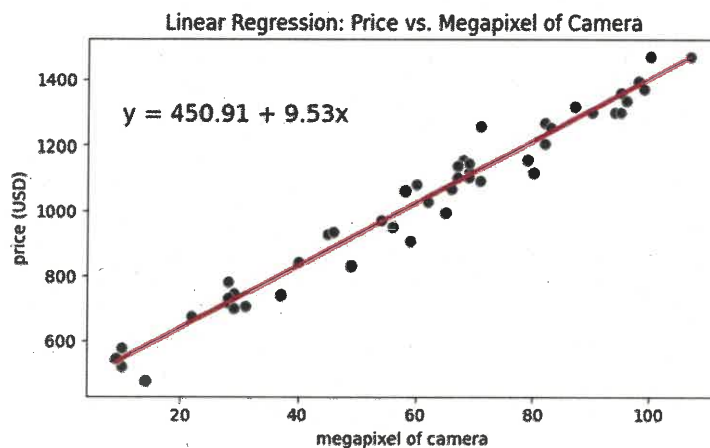
Weights of watermelons (kg)		Unpaired two-sample t-test	
Field A	Field B	Field A	Field B
6.21	7.35	Mean	5.7547.595
5.90	7.72	Variance	0.2240.583
5.04	6.58	Number of watermelons	1414
6.45	6.24	Degrees of freedom (df)	22
6.39	8.13	t-Statistic	-7.666
5.05	7.20	p-value	0.000
5.83	6.94		
5.71	6.78		
5.19	7.65		
5.81	8.50		
5.57	8.26		
5.32	8.00		
5.87	8.28		
6.21	8.69		

- a) The watermelons in Field B are statistically significantly heavier than in Field A.
- b) Because the p-value is < 0.05 , we can reject the null hypothesis.
- c) The mean weights of watermelons differ significantly on the 5% significance level.
- d) There is a 5% chance of making a Type I error, meaning we might incorrectly conclude that the watermelons in Field A and Field B have different mean weights, even if they actually do not.

Regression Analysis

Q21: Consider the graphic below, which shows a simple linear regression of smartphone prices against the camera resolutions of these smartphones in megapixels. Based on the regression line, what is the estimated price of a smartphone with a 48-megapixel camera? Note that the values in the answer options are rounded to two decimal places.

2 points



- a) 875.85
- b) 945.75
- c) 908.35
- d) 894.25

Q22: Consider the equation of a multiple linear regression below. Which statement about this equation is not correct?

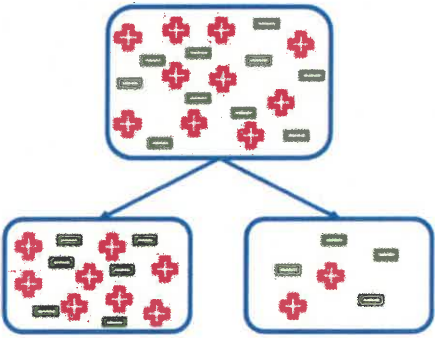
2 points

$$\hat{y} = 200 + 10x_1 + 20x_2$$

Where:

- $\hat{y}$ = estimated smartphone price (in USD),
- x_1 = megapixels of the camera,
- x_2 = battery life (in hours).

- a) For every increase of 4 megapixels, the smartphone price increases by \$40.
- b) For every increase of 5 hours of battery life, the smartphone price increases by \$100.
- c) The estimated price of a smartphone with 0 megapixels and 0 hours battery life is \$200.
- d) A smartphone with a 12-megapixel camera and 25 hours of battery life would cost \$960.

Q23: Which statement about the coefficient of determination (R^2) and the elements of the formula for its calculation is <u>not</u> correct? a) The R^2 represents the proportion of variation in the target variable that is explained by the explanatory variables. b) If SSE equals SST, the value of R^2 will be 1. c) A higher R^2 value indicates a better fit of the model to the data. d) If SSE equals 0, R^2 will be 1, indicating a perfect fit of the model to the data.	2 points
Classification Analysis	
Q24: Consider the following graphic, which shows a classification tree based on 20 observations and two classes represented by plus and minus signs. The classification tree has one split and two terminal nodes. Question: Which of the following statements with regard to the Gini-Impurity is <u>not</u> correct?  a) The Gini-Impurity (left terminal node) can be calculated by: $GI_{\text{left}} = 1 - [(8/14)^2 + (6/14)^2]$. b) The Gini-Impurity (right terminal node) can be calculated by: $GI_{\text{right}} = 1 - [(2/6)^2 + (4/6)^2]$. c) The Gini-Impurity has its maximum if the same number of observations per class is present in a node. d) The Gini-Impurity has its minimum if all observations of a node fall into exactly one class.	2 points

Q25: We trained an image classification model with a total of 10,000 (ten thousand) images to distinguish between cats and dogs. The model was tested with a test dataset and a confusion matrix was created. The number of cat and dog images used, as well as the confusion matrix with the True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN) are listed below. Question: What are the values for Precision, Recall and F1-Score? Note: The values have been rounded to four decimal places and are not given as percentages.

2 points

Number of images:

Total: 10,000

Cats: 4,998

Dogs: 5,002

Confusion matrix:

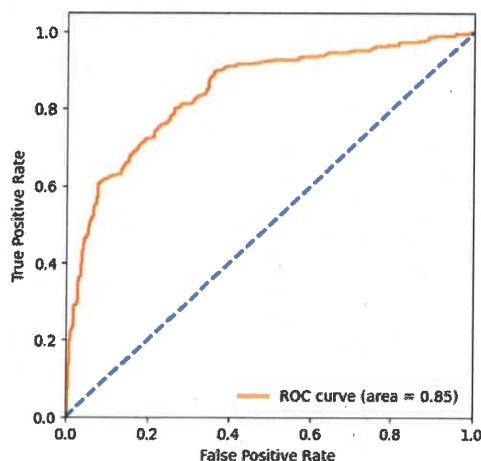
Cats: TP: 4,001 FP: 997

Dogs: FN: 1,250 TN: 3,752

- a) Precision: 0.8005; Recall: 0.7620; F1-Score: 0.7808
- b) Precision: 0.8504; Recall: 0.7802; F1-Score: 0.7402
- c) Precision: 0.7620; Recall: 0.8005; F1-Score: 0.7808
- d) Precision: 0.7802; Recall: 0.8504; F1-Score: 0.7402

Q26: Consider the ROC curve of a binary classification model shown below. Which of the following statements about the ROC curve and the AUC (area under the curve) is not correct?

2 points

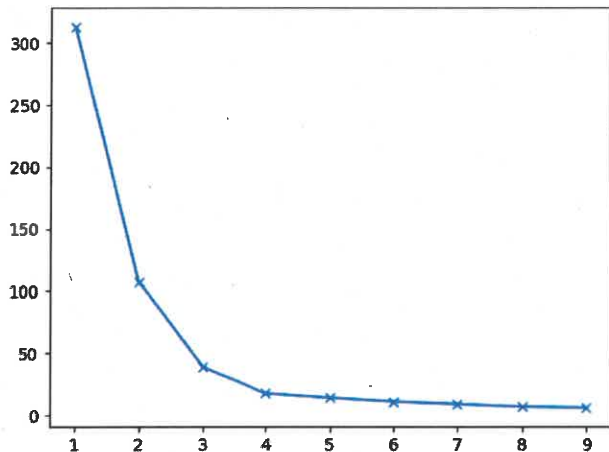


- a) A perfect classification model has an AUC of 1.
- b) A ROC curve shows the model performance of a classification model for different classification thresholds.
- c) An AUC of 0.5 indicates that the model performs no better than random guessing.
- d) Classification models with ROC curves closer to the top-left corner of the plot indicate weaker performance compared to models with curves further from the top-left corner.

Cluster Analysis

Q27: Consider the graph below (with axes labels removed) representing the Elbow method used in k-means clustering. Which statement with regard to the k-means method, the graph shown, and the Elbow method is not correct?

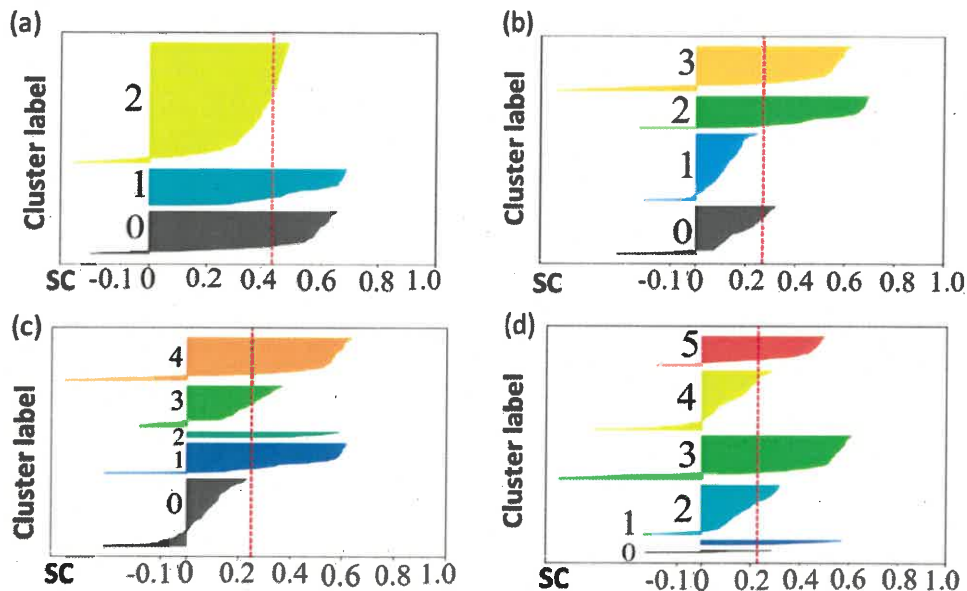
2 points



- a) In the context of k-means, the graph plots the within-cluster sum of squares (WCSS) on the vertical axis against the number of clusters k on the horizontal axis.
- b) The within-cluster sum of squares (WCSS) measures the sum of the squared distances between each data point in a cluster and the cluster centroid. It measures the variation of data points within each cluster.
- c) The elbow method helps determine the optimal number of clusters by identifying the point where adding more clusters does not significantly reduce the within-cluster sum of squares (WCSS), suggesting the ideal k .
- d) The elbow point corresponds to the k value where the within-cluster sum of squares (WCSS) reaches 0, indicating that all data points are perfectly clustered.

Q28: Consider the following Silhouette plots with (a) 3 clusters, (b) 4 clusters, (c) 5 clusters, and (d) 6 clusters. Which statement about the Silhouette method in general and the shown plots in particular is not correct? Note that the terms Silhouette coefficient (SC) and Silhouette score are used interchangeably.

2 points



- a) The Silhouette coefficient measures how similar a point is to its own cluster compared to other clusters and ranges from -1 to 1.
- b) A higher Silhouette coefficient means that the data points are better clustered, and values close to 1 indicate well-separated and compact clusters.
- c) A negative Silhouette coefficient indicates that a data point is assigned to the correct cluster, i.e. it is closer to points in its own cluster than to points in other clusters.
- d) In these graphs, the dashed red line represents the average Silhouette coefficient for all points, and the best clustering solution usually has the highest average Silhouette value.

Text Mining & Sentiment Analysis	
Q29: Take the sentence "Paul wanted a better job." Which of the following options corresponds to the complete preprocessing of the sentence according to the steps for text wrangling and preprocessing presented in class? a) ["paul", "want", "good", "job"] b) ["paul", "want", "a", "good", "job"] c) ["Paul wanted a better job"] d) ["Paul", "wanted", "a", "better", "job"]	2 points
Q30: What is output of the print() function in the Python code below? Note that the 'pos=' parameter relates to the word types <i>verb</i>, <i>noun</i>, and <i>adjective</i>. <pre># Import necessary libraries import nltk from nltk import pos_tag nltk.download('omw-1.4') nltk.download('wordnet') nltk.download('averaged_perceptron_tagger') from nltk.corpus import wordnet from nltk.stem import WordNetLemmatizer # Initialize the lemmatizer lemmatizer = WordNetLemmatizer() # Lemmatize the words list = [] list.append(lemmatizer.lemmatize("wrote", pos="v")) list.append(lemmatizer.lemmatize("children", pos="n")) list.append(lemmatizer.lemmatize("happier", pos="a")) list.append(lemmatizer.lemmatize("running", pos="v")) list.append(lemmatizer.lemmatize("employees", pos="n")) # Print the lemmatized words print(list)</pre> a) ['written', 'child', 'happy', 'run', 'employee'] b) ['wrote', 'children', 'happier', 'run', 'employees'] c) ['writing', 'child', 'happy', 'running', 'employee'] d) ['write', 'child', 'happy', 'run', 'employee']	2 points

END OF EXAM